**Supplementary Material**

# Soil Hydraulic Memory and the Recoverability of Evaporation Savings and the Closure of the Water Balance in a Simulation Study with Daisy

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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## S1. What is new in this article

The table sets each contribution against what was established before it. The derivations that follow mark which equations are taken from the literature and which are new.

**Table S1:** What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

| # | Established before | New in this article | Evidence here |
|---|---|---|---|
| 1 | Drainage and evaporation compete for stored water; their rates follow from the van Genuchten–Mualem functions and from stage-1/stage-2 evaporation theory (Lehmann et al., 2008; Mualem, 1976; Or et al., 2013; van Genuchten, 1980). | *Hydraulic memory*, the share of released stored water that drains, defined as a measurable quantity and predicted for a soil not used in the fit by a one-parameter rate-competition law. | Spans 0.02 to 0.85 over ten soils; ordered by $n$ ($\rho = 0.95$) and $K_s$ ($\rho = 0.97$); leave-one-soil-out RMSE 0.058, below every structural predictor. |
| 2 | Water saved from evaporation is counted as water gained, while system-scale studies warn that savings can be illusory (Grafton et al., 2018; Perry et al., 2017). | A soil- and climate-specific *recoverable share* of an evaporation saving, $\rho_R = 1 - \psi_D$, with the drainage that is not recovered quantified per soil. | No recovery in sand, 0.20 in loamy sand; warming shifts the drainage share by $-0.16$, a wetter climate by $+0.42$; nitrogen (negative control) has no effect. |
| 3 | Equation discovery recovers laws from data (Brunton et al., 2016; Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020); energy- and water-balance closure gaps are known for flux towers (Foken, 2008; Wilson et al., 2002). | A blind invariant search with a permutation null and positive and negative controls, applied to the outputs of a process model, recovering its water balance and exposing an unclosed evaporative term. | Balance coefficients within 5 % of the exact values; an unattributed term of 164 mm $\mathrm{yr}^{-1}$ (30 % of ET); storage non-closure of 16 % in two experiments. |

| # | Established before | New in this article | Evidence here |
|---|---|---|---|
| 4 | Emulators make process models fast but their safe targets are rarely stated (Asher et al., 2015; Razavi et al., 2012). | A map of which flux of the balance can be emulated by a compact law and which cannot. | A three-term ET law equals gradient boosting out of sample ($R^2 = 0.898$ by experiment, 0.904 by soil); drainage resists every law tried. |
| 5 | Savings are judged against the reported soil evaporation. | The two analyses change each other: against the reported soil evaporation alone, the drainage share of an evaporation change is misjudged. | Misjudged in most soils and of the wrong sign in two. |

## S2. The author's contributions: problem, derivation and pinpoints

Each contribution of the article is set out in the same way: *the problem* it answers, stated in a few lines; *how the author developed it*; *the result*, with the parts that matter marked by circled labels (1), (2), …; and *the pinpoints*, which say for each circled mark what it shows and why it is so. Every equation that the author developed is circled in red pencil and labelled *developed by the author*. Numbers come from the data and can be recomputed in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`.

### S2.1 Contribution 1: hydraulic memory as a measurable share

**The problem**

After a wet spring, a soil releases stored water over the season. Does that water feed the crop, or does it drain away? Soil physics says both happen, but gives no single number per soil that can be measured, compared and predicted.

**How the author developed it**

**(a)** Define the storage release of a run, $\bar{R} = ET + D - P_r - I$ (water given away by the store). **(b)** In experiment E9 the starting wetness is varied at fixed rainfall, so $\bar{R}$ varies while the weather does not. **(c)** Within each rainfall level, regress each flux on $\bar{R}$; pool the

levels (Section S3.3). **(d)** Because the four fluxes add to $\bar{R}$ plus a constant, their slopes add to one.

**The result**

$$\underbrace{\phi_x}_{(1)} = \frac{\sum_P \sum_{c\in P} \overbrace{(\bar{R}_c - \bar{\bar{R}}_P)(x_c - \bar{x}_P)}^{(2)}}{\sum_P \sum_{c\in P} (\bar{R}_c - \bar{\bar{R}}_P)^2}$$

$$\underbrace{\phi_{E_s} + \phi_T + \phi_D + \phi_{\text{oth}} = 1}_{(3)}$$

$$\underbrace{\phi_D = 0.02 \text{ to } 0.85}_{(4)}$$

*developed by the author*

**Pinpoints: what each circled mark shows and why**

(1) The memory fate of flux $x$: the share of one released millimetre that leaves by $x$. It is a number between 0 and 1 for each soil, so soils can be ranked and compared.

(2) Deviations are taken *within* each rainfall level $P$, so the effect of the weather is removed and only the effect of the stored water is measured.

(3) The four shares add to one exactly, because a released millimetre must leave by one of them; this is a built-in check of the calculation.

(4) The drained share ranges from 0.02 (silty clay loam: almost all released water is used) to 0.85 (sand: almost all drains). This forty-fold range is the soil's hydraulic memory.

### S2.2 Contribution 2: a one-parameter law for hydraulic memory

**The problem**

Can the drained share of a soil that was never simulated be predicted from one measured property?

## How the author developed it

Stored water leaves by two competing first-order routes, drainage at a rate proportional to $K_s$ and withdrawal by roots and surface at a rate proportional to $q^*$ (Section S3.5); the share taken by drainage is the ratio of its rate to the sum.

## The result

$$\underbrace{\phi_D}_{1} = \frac{\overbrace{K_s}^{2}}{K_s + \underbrace{q^*}_{3}}$$

$$\underbrace{q^* = 71.5\ \mathrm{cm\,d^{-1}}}_{4}$$

$$\underbrace{\mathrm{LOSO\ RMSE} = 0.058}_{5}$$

*developed by the author*

## Pinpoints: what each circled mark shows and why

(1) The predicted memory fate.

(2) The saturated conductivity is the only soil input; it sets how fast released water can drain.

(3) $q^*$ is one constant for all soils: the effective rate of withdrawal by evaporation and roots, expressed as a conductivity.

(4) $q^*$ is two orders of magnitude above the mean evapotranspiration, which shows where the competition is decided: in the wet, fast-draining phase just after the store is filled, not over the season.

(5) Predicting each soil from the other seven, the error was 0.058, smaller than with $n$ (0.100) or any other structural predictor.

### S2.3 Contribution 3: the recoverable share of an evaporation saving

**The problem**

A mulch saves evaporation. Is the saved water gained for the crop or the irrigation account, or is it lost to drainage?

**How the author developed it**

Paired runs that differ only in the evaporation factor (E16) or in the mulch level (E7) were differenced; the water balance gives $\psi_T + \psi_D + \psi_I \approx 1$ for the fates of the evaporation change; the recoverable share is what does not drain.

**The result**

$$\underbrace{\rho_R}_{(1)} = 1 - \underbrace{\psi_D}_{(2)}$$

*developed by the author*

| Soil | $\rho_R$ (evaporation factor, E16) | $\rho_R$ (mulch, E7) |
|---|---|---|
| Sand | (3) −0.21 (none) | 0.29 |
| Loamy sand | 0.20 | 0.31 |
| Loam | 0.53 | 0.48 |
| Silty clay loam | 0.69 | 0.59 |

**Pinpoints: what each circled mark shows and why**

(1) The share of a saved millimetre of evaporation that is recovered (as transpiration or as less irrigation).

(2) The loss fate: the share of the saving that drains instead. It follows the same order as the memory fate of Contribution 1, which links the two experiments.

(3) In sand nothing is recovered: the saving drains (the negative value shows that drainage even exceeds the saving within the uncertainty of the regression). A mulch on sand saves evaporation but not water for the crop.

## S2.4 Contribution 4: discovering the model's water balance without being told

### The problem

Can an automatic method find the conservation law hidden in a process model's outputs, and use it to detect what the outputs leave out?

### How the author developed it

The 20-year fluxes $P_r$, $I$, $ET$ and $D$ of all valid runs were centred and stacked as a matrix; its singular-value decomposition gives the combination of columns closest to zero; a permutation of each column gives the distribution under no structure; the vector was rescaled so that ET has coefficient $-1$.

### The result

$$\underbrace{\mathbf{c}}_{①} = (c_{P_r}, c_I, c_{ET}, c_D) = (\underbrace{1.037}_{②},\ 0.957,\ -1,\ \underbrace{-1.036}_{③})$$

$$\underbrace{R = ET - T - E_s - E_c = 164\ \mathrm{mm\,yr^{-1}}}_{④}$$

*developed by the author*

### Pinpoints: what each circled mark shows and why

① The right singular vector of the flux matrix with the smallest singular value: the linear combination of fluxes that is closest to zero for all 10 041 runs.

② Rain and irrigation enter with $+1$ and evapotranspiration and drainage with $-1$, within 5 %: the method recovered $P_r + I - ET - D \approx 0$, the water balance, without being given it.

③ The small departures from $\pm 1$ measure what the four fluxes do not contain (storage change and runoff), which is how the closure residual was found.

④ Evapotranspiration minus its reported parts is 164 mm a year, 30 % of ET. The full outputs show this is evaporation of ponded water (21.6 % of ET), litter (7.6 %) and snow (2.7 %), which the run table did not report.

### S2.5 Contribution 5: a compact law for evapotranspiration

**How the author developed it**

Log and dimensionless candidate terms were added one at a time by forward selection; a term was kept only if the cross-validated $R^2$ (experiments or soils left out) rose by more than 0.003; the selected law was compared with gradient boosting on the same folds.

**The result**

$$\ln ET = 2.503 + \underbrace{0.646 \ln W}_{(1)} \underbrace{-0.299\, n}_{(2)} + \underbrace{0.029 \ln(1+I)}_{(3)}$$

$$\underbrace{R^2 = 0.898 \text{ and } 0.904}_{(4)}$$

*developed by the author*

**Pinpoints: what each circled mark shows and why**

(1) The elasticity of ET to the water input $W = P_r + I$ is 0.65, less than one: a part of every extra millimetre drains instead of evaporating.

(2) A larger $n$ (coarser soil) lowers ET at the same input: coarse soils lose more to drainage, the same mechanism as Contribution 1.

(3) Irrigation has a small effect of its own beyond its part of $W$: it is applied when the root zone is dry, at the time the crop can use it; the coefficient is small, so most of the effect of irrigation is already carried by $W$.

(4) Out-of-sample skill on experiments and on soils left out, equal to gradient boosting: ET can be emulated by three terms, while drainage could not be emulated by any law tried.

## S3. A guide to the results with worked numbers

The results of the article can be followed with a calculator or with the workbook (sheet `EQ_P1`). Each part gives the idea, the formula and a worked example.

### S3.1 1. the water balance of a soil column

Over a year, water in (rain $P$ and irrigation $I$) equals water out (evapotranspiration *ET*, drainage $D$ below 1 m, runoff $Q$) plus the change of water stored, $\Delta S$: $P + I = ET + D + \Delta S + Q$. *Example.* Coarse loam without irrigation: $P = 464.4$, $ET = 466.8$, $D = 6.1$ mm; $ET + D - P = 8.5$ mm came from storage.

### S3.2 2. the storage release

$\bar{R} = ET + D - P - I$ is the water the soil gave away from its store. Starting the run wetter makes $\bar{R}$ larger; the experiment E9 does this on purpose with ten starting suctions.

### S3.3 3. the memory fate

Plot drainage against $\bar{R}$ for the runs with the same rainfall: the slope is the share of each released millimetre that drained, $\phi_D$. The same slope for transpiration, soil and other evaporation gives $\phi_T$, $\phi_{E_s}$, $\phi_{\text{oth}}$, and the four add to 1 because the released water must leave by one of them. *Example (three runs).* If $\bar{R} = 10$, 20, 30 mm and $D = 1$, 4, 7 mm, the slope is $(7 - 1)/(30 - 10) = 0.3$: 30 % of released water drains.

### S3.4 4. why drainage wins in coarse soils, the rate law

Released water leaves by two routes that compete: drainage, fast when the saturated conductivity $K_s$ is large, and withdrawal by roots and the surface, with a fixed rate $q^*$. The share that drains is $\phi_D = K_s/(K_s + q^*)$, with $q^* = 71.5\,\text{cm}\,\text{d}^{-1}$. *Example.* Sand, $K_s = 712.8$: $712.8/(712.8 + 71.5) = 0.91$ (observed 0.85). Silty clay loam, $K_s = 1.68$: $1.68/73.18 = 0.023$ (observed 0.022).

### S3.5 5. the recoverable share of an evaporation saving

When a mulch or a lower evaporation demand saves evaporation, the saved water goes to transpiration, to less irrigation, or to drainage. Only the part that does not drain is recovered: $\rho_R = 1 - \psi_D$. *Example.* Loamy sand, $\psi_D = 0.80$ in the edge-case experiment: $\rho_R = 0.20$; only a fifth of a saving is recovered.

### S3.6 6. checking the model's own bookkeeping

If the outputs of the model obey the balance of Part 1, the columns $P$, $I$, *ET* and $D$ of all runs are linked by one linear relation. The singular-value decomposition finds the combination of columns that is closest to zero; it gave coefficients within 5 % of $(1, 1, -1, -1)$, so the method finds the balance without being told it.

### S3.7 7. the unattributed evaporation

*ET* minus transpiration, soil evaporation and canopy evaporation should be zero; in the run table it averaged 164 mm a year (30 % of *ET*). The full outputs show it is evaporation of ponded water, litter and snow, which the run table did not report separately.

### S3.8 8. reproduce everything

Sheet `EQ_P1` computes Part 4, Part 5 and the storage release of each soil from the runs; the scripts of `Manuscripts/reproducibility` regenerate every number of the article.

## S4. Derivations of the equations

Sections S2.1, S2.2 and S2.8 re-derive established results (Mualem, 1976; van Genuchten, 1980) and standard statistical tools for completeness. The memory fate (S2.3), the loss fates and the recoverable share (S2.4), the rate-competition law of hydraulic memory (S2.5), the invariant search applied to a process model (S2.6), the closure residual and the unattributed term (S2.7) and the compensation coefficient (S2.9) are definitions and results of this article.

### S4.1 The retention and conductivity functions

With the capillary law $r \propto 1/h$, Mualem (1976) expressed the relative conductivity of a bundle of connected capillaries as

$$K_r(S_e) = S_e^{\lambda} \left[ \int_0^{S_e} h(x)^{-1} \mathrm{d}x \Big/ \int_0^{1} h(x)^{-1} \mathrm{d}x \right]^2 . \tag{S1}$$

Inserting the inverse of the van Genuchten curve, $h(x) = \alpha^{-1}(x^{-1/m} - 1)^{1/n}$, and substituting $y = x^{1/m}$ gives $\int_0^{S_e} h^{-1}\mathrm{d}x = \alpha m \int_0^{S_e^{1/m}} y^{m-1+1/n}(1-y)^{-1/n}\mathrm{d}y$. With $m = 1 - 1/n$ the power

of $y$ vanishes and the integral is $\alpha[1-(1-S_e^{1/m})^m]$; dividing by its value at $S_e=1$ ($\alpha$) and squaring yields the Mualem conductivity of the paper, $K=K_sS_e^{\lambda}[1-(1-S_e^{1/m})^m]^2$. For large $\alpha h$, $S_e \simeq (\alpha h)^{-(n-1)}$ and $K_r \simeq m^2(\alpha h)^{-[2n+\lambda(n-1)]}$: the conductivity of a soil with large $n$ falls faster with suction, which is the property that orders hydraulic memory.

### S4.2 The water balance and the storage release

Over a period, mass conservation of the soil column with surface runoff $Q_r$ is $P_r+I=ET+D+\Delta S+Q_r$. Rearranging,

$$\bar{R} \equiv ET + D - P_r - I = -(\Delta S + Q_r), \tag{S2}$$

so the storage release is positive when the column loses stored water (or sheds no runoff) and supplies the fluxes. In the initial-moisture experiment, a wetter initial state raises the stored water that is released over the run, which is the variation used to measure the memory fates.

### S4.3 The memory fate as a pooled within-group slope

Within a group of runs with the same rainfall scale $P$, write each flux as $x_c = a_P + \phi_x \bar{R}_c + e_c$. Minimising $\sum_P \sum_{c\in P} e_c^2$ over the group intercepts $a_P$ gives $a_P = \bar{x}_P - \phi_x \overline{\bar{R}}_P$, and then over $\phi_x$

$$\phi_x = \frac{\sum_P \sum_{c\in P} (\bar{R}_c - \overline{\bar{R}}_P)(x_c - \bar{x}_P)}{\sum_P \sum_{c\in P} (\bar{R}_c - \overline{\bar{R}}_P)^2}. \tag{S3}$$

Because the four fluxes add to $ET+D=\bar{R}+P_r+I$ and $P_r+I$ is constant within a group, the deviations of the four fluxes add to the deviation of $\bar{R}$, and the four slopes add to one: $\phi_{E_s}+\phi_T+\phi_D+\phi_{\text{oth}}=1$. Each $\phi_x$ is therefore the share of a released millimetre that leaves by flux $x$.

### S4.4 Loss fates and the recoverable share

Between two levels of the evaporation factor the change of the non-transpirational evapotranspiration $\delta ET_{\text{nt}}$ must be balanced by the other fluxes. With storage changes negligible over 20 years, $\delta ET_{\text{nt}} + \delta T + \delta D - \delta I \simeq 0$, hence

$$\psi_T + \psi_D + \psi_I \simeq 1, \qquad \psi_x = \mp \frac{\delta x}{\delta ET_{\text{nt}}}, \tag{S4}$$

the signs chosen so that each $\psi$ is positive when the flux compensates the evaporation change. Of a millimetre of evaporation saved, $\psi_D$ leaves as drainage and is not recovered by the crop or the irrigation account; the recoverable share is $\rho_R = 1 - \psi_D$.

### S4.5 The rate-competition law

Let stored water $S$ above field capacity leave by two parallel first-order sinks, drainage with rate constant $k_D \propto K_s/L$ and withdrawal by evaporation and uptake with $k_W \propto q^*/L$, where $L$ is a common length. Then $\mathrm{d}S/\mathrm{d}t = -(k_D + k_W)S$ and the drainage leaving over the release is

$$\int_0^\infty k_D S \,\mathrm{d}t = \frac{k_D}{k_D + k_W} S_0 \quad \Longrightarrow \quad \phi_D = \frac{K_s}{K_s + q^*}. \tag{S5}$$

The law is bounded in $[0, 1]$, tends to 0 for an impermeable soil and to 1 for a free-draining one, and is the only one-parameter form with these limits that is linear in $K_s$ at small $K_s$. By the Buckingham theorem, $\phi_D = F(K_s/q^*, n, \alpha h^*)$; the law keeps the first group only.

### S4.6 The invariant search by singular-value decomposition

Arrange the 20-year mean fluxes of all runs as the rows of a matrix $\mathbf{F}$ (columns $P_r, I, \mathit{ET}, D$, centred). A linear invariant is a vector $\mathbf{c}$ with $\mathbf{Fc} \approx \mathbf{0}$. The best such vector in the least-squares sense, with $\|\mathbf{c}\| = 1$, minimises $\|\mathbf{Fc}\|^2 = \mathbf{c}^\top \mathbf{F}^\top \mathbf{F}\mathbf{c}$, which is the eigenvector of $\mathbf{F}^\top\mathbf{F}$ with the smallest eigenvalue, that is the right singular vector of $\mathbf{F}$ with the smallest singular value. Rescaled so that the *ET* coefficient is $-1$, it is the discovered balance; the permutation null shuffles each column independently and so destroys any true invariant, which gives the distribution of the smallest singular value under no structure.

### S4.7 The closure residual and the unattributed term

With $W = P_r + I$, the residual $\varepsilon = (W - \mathit{ET} - D)/W \simeq (\Delta S + Q_r)/W$ is the share of the input not leaving by evapotranspiration or drainage. The unattributed evapotranspiration is $R = \mathit{ET} - T - E_s - E_c$; in the full outputs it equals the evaporation of ponded water, litter and snow (data folder, closure within 1.1 mm).

### S4.8 Log-ratio coordinates and the sparse law

Fluxes that share a total are compositional; the centred log-ratio $\mathrm{clr}(\mathbf{x})_i = \ln x_i - \frac{1}{5}\sum_j \ln x_j$ removes the constraint. The sparse laws regress $\ln y$ on a small set of log or dimensionless candidates chosen by forward selection; out-of-sample skill is $R^2_{\mathrm{cv}} = 1 - \sum(\ln y_i - \widehat{\ln y_i}^{(-g)})^2 / \sum(\ln y_i - \overline{\ln y})^2$, with each prediction made by a model fitted without the group $g$ (experiment or soil) of run $i$.

### S4.9 Irrigation compensation

With the same balance, a change of rainfall $\Delta P_r$ at fixed trigger is offset partly by irrigation: $\kappa = -\Delta I / \Delta P_r$. When drainage and evapotranspiration do not change, $\kappa = 1$; when part of the extra rain drains, $\kappa < 1$.

## S5. Why the results happen

*Why hydraulic memory differs between soils.* Water released from storage leaves by the faster of two routes. In coarse soils (large $n$, high $K_s$) the conductivity at field capacity is high, so released water drains before roots and the surface can use it; in fine soils the conductivity is low and the same water stays near the roots and the surface, where it is transpired or evaporated.

*Why the memory transfers to other experiments.* The same competition decides where any extra water goes, whether it comes from initial storage, from a lower evaporation factor or from a mulch; the share is a property of the soil's conductivity, not of the source of the water.

*Why warming raises and wetter climates lower the recoverable share.* Warming raises the evaporative demand, so the withdrawal side of the competition is faster and less of a saved millimetre drains. A wetter climate keeps the profile closer to field capacity, where drainage is fast, so more of a saving drains.

*Why nitrogen has no effect.* The nitrogen rate changes the crop's growth but not the hydraulic properties, and in these runs it changes the water fluxes too little to move the shares; it therefore serves as a negative control.

*Why an evaporative term is unattributed.* The reported components cover transpiration, soil evaporation and canopy evaporation; evaporation of ponded water, litter and snow is part of evapotran-

spiration but not of those components, so it appears as a residual in the run table.

*Why evapotranspiration follows a compact law and drainage does not.* Evapotranspiration is limited by energy and by water and is smooth in its inputs over 20 years. Drainage is a threshold process (water drains only above field capacity) whose annual total depends on the timing of rain and irrigation events, which the long-term means do not contain.

*Why judging a saving against soil evaporation alone misleads.* A mulch changes soil evaporation and also the unattributed evaporation; if only the reported soil evaporation is used as the denominator, the drainage share of the change is computed against the wrong total and can change sign.

## S6. The data

Table S2 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analyses of the paper, printed in full or up to 220 rows; each is also in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

**Table S2:** The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; $Y$ yield (Mg ha$^{-1}$), $I$ irrigation, $D$ drainage at 1 m and $ET$ (mm yr$^{-1}$), $N_L$ nitrate leached (kg N ha$^{-1}$ yr$^{-1}$), $t$ run time of Daisy (s). Every row is one file of the folder, linked.

| Exp. | File | Runs | Valid | Factors varied | $Y$ | $I$ | $D$ | $ET$ | $N_L$ | $t$ |
|---|---|---|---|---|---|---|---|---|---|---|
| E0 | `Baselines` | 10 | 10 | soil | 2.19 | 0 | 55 | 418 | 158 | 231 |
| E1 | `Doses` | 200 | 189 | soil, rate, dose | 5.13 | 198 | 98 | 564 | 79 | 195 |
| E2 | `TriggerPressure` | 200 | 197 | soil, trigger | 4.70 | 169 | 80 | 559 | 90 | 139 |
| E3 | `SensorDepth` | 200 | 200 | soil, sensor depth | 4.94 | 199 | 86 | 583 | 86 | 146 |
| E4 | `Methods` | 100 | 96 | soil, method | 5.01 | 164 | 88 | 542 | 83 | 147 |
| E5 | `SHP Dynamics` | 300 | 300 | soil, method, tillage | 4.13 | 152 | 43 | 579 | 84 | 189 |
| E6 | `Mediterranean Pollution` | 210 | 193 | soil, method, trigger, N rate | 3.64 | 84 | 239 | 620 | 62 | 118 |
| E7 | `Mulching` | 200 | 200 | soil, mulch | 5.29 | 183 | 96 | 555 | 74 | 161 |
| E8 | `Rain Intensity Moisture` | 1000 | 1000 | initial suction, rain scale | 0.68 | 0 | -1 | 282 | -0 | 6 |
| E9 | `Soil Moisture Rain` | 1000 | 987 | soil, initial suction, rain scale | 0.75 | 0 | 19 | 247 | 33 | 7 |
| E10 | `Amount Timing Soil` | 1000 | 954 | soil, rate, dose, window | 4.42 | 139 | 76 | 532 | 97 | 146 |
| E11 | `Temp Evap Rain` | 1000 | 1000 | rain scale, warming, demand | 3.81 | 324 | 18 | 754 | 66 | 151 |
| E12 | `Sensor Trigger Rain` | 1000 | 1000 | trigger, sensor depth, rain scale | 5.88 | 260 | 35 | 670 | 32 | 155 |
| E13 | `N Water Soil` | 1000 | 975 | soil, method, rate, dose, trigger, N rate | 4.00 | 162 | 83 | 547 | 79 | 207 |
| E14 | `Mulch Rain` | 1000 | 1000 | rain scale, mulch | 6.15 | 235 | 43 | 634 | 25 | 173 |
| E15 | `Method Depth Amount` | 1000 | 1000 | method, rate, dose, root depth | 6.02 | 217 | 61 | 619 | 37 | 157 |
| E16 | `Edge Cases` | 770 | 740 | soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand | 2.17 | 344 | 185 | 600 | 93 | 224 |

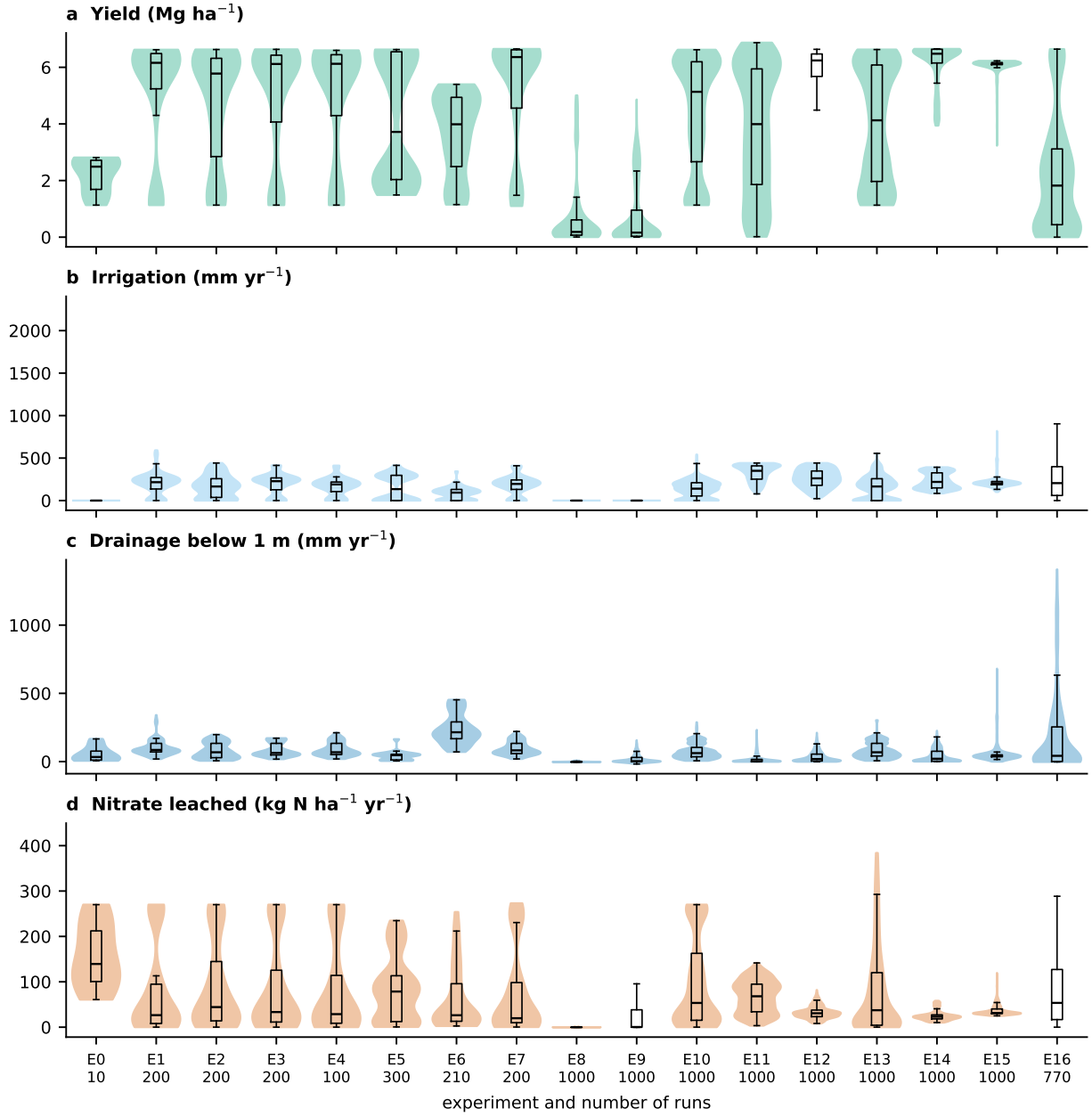



**Figure S1:** Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

## p1_audit_link.json

Source: `Manuscripts/results/p1_audit_link.json`; 46 values. Workbook sheet `R_p1_audit_link`.

| quantity | value |
|---|---|
| E7.S01.psiD_Ent | 0.2975 |
| E7.S01.psiD_Es | 0.3089 |
| E7.S01.share_of_dEnt_in_Es | 0.5618 |
| E7.S01.n | 20 |
| E7.S02.psiD_Ent | 0.6073 |
| E7.S02.psiD_Es | 0.3798 |
| E7.S02.share_of_dEnt_in_Es | 0.4582 |

| quantity | value |
|---|---|
| E7.S02.n | 20 |
| E7.S03.psiD_Ent | 0.7062 |
| E7.S03.psiD_Es | -0.9307 |
| E7.S03.share_of_dEnt_in_Es | -0.3733 |
| E7.S03.n | 20 |
| E7.S04.psiD_Ent | 0.6867 |
| E7.S04.psiD_Es | -0.3497 |
| E7.S04.share_of_dEnt_in_Es | -0.3967 |
| E7.S04.n | 20 |
| E7.S05.psiD_Ent | 0.6628 |
| E7.S05.psiD_Es | 0.157 |
| E7.S05.share_of_dEnt_in_Es | 0.06688 |
| E7.S05.n | 20 |
| E7.S06.psiD_Ent | 0.5212 |
| E7.S06.psiD_Es | 0.4076 |
| E7.S06.share_of_dEnt_in_Es | 0.4787 |
| E7.S06.n | 20 |
| E7.S07.psiD_Ent | 0.3955 |
| E7.S07.psiD_Es | 0.4209 |
| E7.S07.share_of_dEnt_in_Es | 0.5235 |
| E7.S07.n | 20 |
| E7.S08.psiD_Ent | 0.5489 |
| E7.S08.psiD_Es | 0.359 |
| E7.S08.share_of_dEnt_in_Es | 0.3922 |
| E7.S08.n | 20 |
| E7.S09.psiD_Ent | 0.4257 |
| E7.S09.psiD_Es | 0.441 |
| E7.S09.share_of_dEnt_in_Es | 0.5036 |
| E7.S09.n | 20 |
| E7.S10.psiD_Ent | 0.4137 |
| E7.S10.psiD_Es | 0.4541 |
| E7.S10.share_of_dEnt_in_Es | 0.4989 |
| E7.S10.n | 20 |
| E7_summary.median_share_of_dEnt_in_Es | 0.4684 |
| E7_summary.median_abs_diff_psiD | 0.1517 |
| E7_summary.max_abs_diff_psiD | 1.637 |
| E7_summary.n_soils | 10 |
| E7_summary.n_soils_diff_gt_0p1 | 6 |
| E7_summary.n_soils_sign_flip | 2 |

## p1_core.json

Source: `Manuscripts/results/p1_core.json`; 270 values, first 220 shown. Workbook sheet `R_p1_core`.

| quantity | value |
|---|---|
| memory.S01.fE | 0.1451 |
| memory.S01.fT | 0.75 |
| memory.S01.fD | 0.02912 |
| memory.S01.fO | 0.07576 |
| memory.S01.ci.soil_evap_mm | 0.113, 0.2002 |
| memory.S01.ci.transpiration_mm | 0.7102, 0.7721 |
| memory.S01.ci.drainage_1m_mm | 0.02449, 0.03256 |
| memory.S01.ci.oth | 0.06013, 0.09077 |
| memory.S01.capFC | 82.79 |
| memory.S01.cap60 | 112.6 |
| memory.S01.n | 100 |
| memory.S02.fE | 0.1927 |
| memory.S02.fT | 0.4187 |
| memory.S02.fD | 0.265 |
| memory.S02.fO | 0.1236 |
| memory.S02.ci.soil_evap_mm | 0.1604, 0.2324 |
| memory.S02.ci.transpiration_mm | 0.4009, 0.4444 |
| memory.S02.ci.drainage_1m_mm | 0.1982, 0.313 |
| memory.S02.ci.oth | 0.1086, 0.1398 |
| memory.S02.capFC | 104 |
| memory.S02.cap60 | 145.7 |
| memory.S02.n | 100 |
| memory.S03.fE | 0.07403 |
| memory.S03.fT | 0.07982 |
| memory.S03.fD | 0.8455 |
| memory.S03.fO | 0.0006454 |
| memory.S03.ci.soil_evap_mm | 0.06681, 0.08221 |
| memory.S03.ci.transpiration_mm | 0.04333, 0.1216 |
| memory.S03.ci.drainage_1m_mm | 0.7959, 0.89 |
| memory.S03.ci.oth | 0.0002684, 0.001096 |
| memory.S03.capFC | 4.172 |
| memory.S03.cap60 | 9.87 |
| memory.S03.n | 87 |
| memory.S04.fE | 0.09992 |
| memory.S04.fT | 0.1535 |
| memory.S04.fD | 0.7409 |
| memory.S04.fO | 0.005727 |
| memory.S04.ci.soil_evap_mm | 0.093, 0.1086 |
| memory.S04.ci.transpiration_mm | 0.1329, 0.1871 |
| memory.S04.ci.drainage_1m_mm | 0.7008, 0.7655 |
| memory.S04.ci.oth | 0.004453, 0.007196 |
| memory.S04.capFC | 13.19 |
| memory.S04.cap60 | 25.76 |
| memory.S04.n | 100 |
| memory.S05.fE | 0.1277 |
| memory.S05.fT | 0.2043 |
| memory.S05.fD | 0.643 |
| memory.S05.fO | 0.02492 |

| quantity | value |
|---|---|
| memory.S05.ci.soil_evap_mm | 0.1174, 0.1419 |
| memory.S05.ci.transpiration_mm | 0.1839, 0.2358 |
| memory.S05.ci.drainage_1m_mm | 0.5987, 0.6723 |
| memory.S05.ci.oth | 0.02044, 0.0298 |
| memory.S05.capFC | 41.33 |
| memory.S05.cap60 | 71.37 |
| memory.S05.n | 100 |
| memory.S06.fE | 0.04019 |
| memory.S06.fT | 0.5825 |
| memory.S06.fD | 0.2892 |
| memory.S06.fO | 0.08811 |
| memory.S06.ci.soil_evap_mm | 0.02569, 0.05774 |
| memory.S06.ci.transpiration_mm | 0.554, 0.6231 |
| memory.S06.ci.drainage_1m_mm | 0.239, 0.3242 |
| memory.S06.ci.oth | 0.07656, 0.1007 |
| memory.S06.capFC | 131.8 |
| memory.S06.cap60 | 183.1 |
| memory.S06.n | 100 |
| memory.S07.fE | 0.04151 |
| memory.S07.fT | 0.7155 |
| memory.S07.fD | 0.1473 |
| memory.S07.fO | 0.09566 |
| memory.S07.ci.soil_evap_mm | 0.0304, 0.05195 |
| memory.S07.ci.transpiration_mm | 0.6886, 0.7481 |
| memory.S07.ci.drainage_1m_mm | 0.1124, 0.1754 |
| memory.S07.ci.oth | 0.0862, 0.1052 |
| memory.S07.capFC | 208.1 |
| memory.S07.cap60 | 248.9 |
| memory.S07.n | 100 |
| memory.S08.fE | 0.07846 |
| memory.S08.fT | 0.5547 |
| memory.S08.fD | 0.2852 |
| memory.S08.fO | 0.08163 |
| memory.S08.ci.soil_evap_mm | 0.06028, 0.1076 |
| memory.S08.ci.transpiration_mm | 0.5128, 0.593 |
| memory.S08.ci.drainage_1m_mm | 0.2427, 0.3188 |
| memory.S08.ci.oth | 0.07191, 0.09251 |
| memory.S08.capFC | 57.04 |
| memory.S08.cap60 | 84.54 |
| memory.S08.n | 100 |
| memory.S09.fE | 0.02717 |
| memory.S09.fT | 0.7744 |
| memory.S09.fD | 0.08702 |
| memory.S09.fO | 0.1114 |
| memory.S09.ci.soil_evap_mm | 0.01397, 0.03822 |
| memory.S09.ci.transpiration_mm | 0.7541, 0.7978 |
| memory.S09.ci.drainage_1m_mm | 0.06404, 0.1082 |
| memory.S09.ci.oth | 0.1008, 0.1222 |

| quantity | value |
|---|---|
| memory.S09.capFC | 167.5 |
| memory.S09.cap60 | 194.4 |
| memory.S09.n | 100 |
| memory.S10.fE | 0.02489 |
| memory.S10.fT | 0.8416 |
| memory.S10.fD | 0.02197 |
| memory.S10.fO | 0.1115 |
| memory.S10.ci.soil_evap_mm | 0.01344, 0.0364 |
| memory.S10.ci.transpiration_mm | 0.829, 0.8548 |
| memory.S10.ci.drainage_1m_mm | 0.01336, 0.03072 |
| memory.S10.ci.oth | 0.1015, 0.1211 |
| memory.S10.capFC | 173 |
| memory.S10.cap60 | 190.1 |
| memory.S10.n | 100 |
| rec16.S01.psiD | 0.2946 |
| rec16.S01.psiI | 0.6189 |
| rec16.S01.psiT | 0.1113 |
| rec16.S01.psiS | 0.02473 |
| rec16.S01.dE | 218.5 |
| rec16.S01.ci.psiD | 0.1943, 0.3918 |
| rec16.S01.ci.psiI | 0.448, 0.7595 |
| rec16.S01.ci.psiT | -0.04115, 0.3197 |
| rec16.S01.ci.psiS | 0.00506, 0.04393 |
| rec16.S01.ci.dE | 179.2, 257.9 |
| rec16.S01.n | 32 |
| rec16.S02.psiD | 0.193 |
| rec16.S02.psiI | 0.6578 |
| rec16.S02.psiT | 0.157 |
| rec16.S02.psiS | 0.007809 |
| rec16.S02.dE | 143.6 |
| rec16.S02.ci.psiD | 0.1389, 0.2439 |
| rec16.S02.ci.psiI | 0.3244, 0.858 |
| rec16.S02.ci.psiT | -0.02457, 0.4575 |
| rec16.S02.ci.psiS | 0.004907, 0.0132 |
| rec16.S02.ci.dE | 99.58, 201.3 |
| rec16.S02.n | 13 |
| rec16.S03.psiD | 1.21 |
| rec16.S03.psiI | -0.3839 |
| rec16.S03.psiT | 0.1785 |
| rec16.S03.psiS | 0.004424 |
| rec16.S03.dE | 99.64 |
| rec16.S03.ci.psiD | 0.955, 1.46 |
| rec16.S03.ci.psiI | -0.8056, -0.05112 |
| rec16.S03.ci.psiT | 0.01523, 0.4111 |
| rec16.S03.ci.psiS | -0.001305, 0.01098 |
| rec16.S03.ci.dE | 86.72, 112.8 |
| rec16.S03.n | 32 |
| rec16.S04.psiD | 0.7993 |

| quantity | value |
|---|---|
| rec16.S04.psiI | 0.2189 |
| rec16.S04.psiT | -0.007351 |
| rec16.S04.psiS | 0.01086 |
| rec16.S04.dE | 115 |
| rec16.S04.ci.psiD | 0.6691, 0.9158 |
| rec16.S04.ci.psiI | 0.1352, 0.3027 |
| rec16.S04.ci.psiT | -0.1054, 0.1089 |
| rec16.S04.ci.psiS | 0.005401, 0.01805 |
| rec16.S04.ci.dE | 99.12, 129.1 |
| rec16.S04.n | 32 |
| rec16.S05.psiD | 0.5821 |
| rec16.S05.psiI | 0.4146 |
| rec16.S05.psiT | 0.01543 |
| rec16.S05.psiS | 0.0121 |
| rec16.S05.dE | 141.1 |
| rec16.S05.ci.psiD | 0.4566, 0.6966 |
| rec16.S05.ci.psiI | 0.2391, 0.5755 |
| rec16.S05.ci.psiT | -0.1053, 0.177 |
| rec16.S05.ci.psiS | 0.003533, 0.02239 |
| rec16.S05.ci.dE | 115.9, 167.7 |
| rec16.S05.n | 32 |
| rec16.S06.psiD | 0.469 |
| rec16.S06.psiI | 0.4334 |
| rec16.S06.psiT | 0.1068 |
| rec16.S06.psiS | 0.009196 |
| rec16.S06.dE | 191.5 |
| rec16.S06.ci.psiD | 0.3626, 0.5992 |
| rec16.S06.ci.psiI | 0.1834, 0.607 |
| rec16.S06.ci.psiT | -0.0311, 0.3117 |
| rec16.S06.ci.psiS | -0.003642, 0.02676 |
| rec16.S06.ci.dE | 139.4, 247.3 |
| rec16.S06.n | 32 |
| rec16.S07.psiD | 0.3625 |
| rec16.S07.psiI | 0.5278 |
| rec16.S07.psiT | 0.1168 |
| rec16.S07.psiS | 0.007042 |
| rec16.S07.dE | 217.6 |
| rec16.S07.ci.psiD | 0.2597, 0.4852 |
| rec16.S07.ci.psiI | 0.2777, 0.7039 |
| rec16.S07.ci.psiT | -0.03777, 0.348 |
| rec16.S07.ci.psiS | -0.009443, 0.02719 |
| rec16.S07.ci.dE | 168.3, 271.5 |
| rec16.S07.n | 32 |
| rec16.S08.psiD | 0.533 |
| rec16.S08.psiI | 0.3857 |
| rec16.S08.psiT | 0.09111 |
| rec16.S08.psiS | 0.009891 |
| rec16.S08.dE | 152.5 |

| quantity | value |
|---|---|
| rec16.S08.ci.psiD | 0.4093, 0.6608 |
| rec16.S08.ci.psiI | 0.1227, 0.5953 |
| rec16.S08.ci.psiT | -0.07816, 0.3433 |
| rec16.S08.ci.psiS | -0.005323, 0.02717 |
| rec16.S08.ci.dE | 114.9, 190.4 |
| rec16.S08.n | 32 |
| rec16.S09.psiD | 0.3117 |
| rec16.S09.psiI | 0.645 |
| rec16.S09.psiT | 0.04777 |
| rec16.S09.psiS | 0.004532 |
| rec16.S09.dE | 217.4 |
| rec16.S09.ci.psiD | 0.1979, 0.4445 |
| rec16.S09.ci.psiI | 0.3863, 0.806 |
| rec16.S09.ci.psiT | -0.07744, 0.2568 |
| rec16.S09.ci.psiS | -0.01091, 0.02484 |
| rec16.S09.ci.dE | 165.6, 267.7 |
| rec16.S09.n | 32 |
| rec16.S10.psiD | 0.3103 |
| rec16.S10.psiI | 0.838 |
| rec16.S10.psiT | -0.01707 |
| rec16.S10.psiS | 0.1312 |
| rec16.S10.dE | 222.1 |
| rec16.S10.ci.psiD | 0.217, 0.4126 |
| rec16.S10.ci.psiI | 0.6104, 1.033 |
| rec16.S10.ci.psiT | -0.1507, 0.1387 |
| rec16.S10.ci.psiS | 0.05407, 0.1948 |
| rec16.S10.ci.dE | 176.8, 272.6 |
| rec16.S10.n | 32 |

## p1_ens_structure.json

Source: `Manuscripts/results/p1_ens_structure.json`; 50 values. Workbook sheet `R_p1_ens_structure`.

| quantity | value |
|---|---|
| dE.S01 | 54.3 |
| dE.S02 | 28.7 |
| dE.S03 | 3.4 |
| dE.S04 | 6.9 |
| dE.S05 | 17 |
| dE.S06 | 32.8 |
| dE.S07 | 45.3 |
| dE.S08 | 25.5 |
| dE.S09 | 41.2 |
| dE.S10 | 44.8 |

| quantity | value |
|---|---|
| dD.S01 | 16.5 |
| dD.S02 | 20.8 |
| dD.S03 | 6.1 |
| dD.S04 | 7.9 |
| dD.S05 | 9.5 |
| dD.S06 | 19.5 |
| dD.S07 | 25.4 |
| dD.S08 | 15.4 |
| dD.S09 | 25.8 |
| dD.S10 | 27.1 |
| dI.S01 | 46.6 |
| dI.S02 | 10.4 |
| dI.S03 | 0 |
| dI.S04 | -0.4 |
| dI.S05 | -5.5 |
| dI.S06 | 18.1 |
| dI.S07 | 45.1 |
| dI.S08 | 9.2 |
| dI.S09 | 42.8 |
| dI.S10 | 49.1 |
| tillage_offset.S01 | 0.27 |
| tillage_offset.S02 | -1.82 |
| tillage_offset.S03 | 5.34 |
| tillage_offset.S04 | 3.69 |
| tillage_offset.S05 | 1.76 |
| tillage_offset.S06 | -1.32 |
| tillage_offset.S07 | -0.25 |
| tillage_offset.S08 | -3.69 |
| tillage_offset.S09 | -2.79 |
| tillage_offset.S10 | -1.7 |
| tc_effect.S01 | 0 |
| tc_effect.S02 | 0 |
| tc_effect.S03 | 0 |
| tc_effect.S04 | 0 |
| tc_effect.S05 | 0 |
| tc_effect.S06 | 1.09 |
| tc_effect.S07 | 0.23 |
| tc_effect.S08 | 2.54 |
| tc_effect.S09 | 2.06 |
| tc_effect.S10 | 1.48 |

## p1_loso_summary.json

Source: `Manuscripts/results/p1_loso_summary.json`; 5 values. Workbook sheet `R_p1_loso_summary`.

| quantity | value |
|---|---|
| TAW | 0.1825 |
| $S_e$(100) | 0.1358 |
| $n$ | 0.09716 |
| $/log K_s$* | 0.05811 |
| mean only | 0.3178 |

## p1_mech.json

Source: `Manuscripts/results/p1_mech.json`; 5 values. Workbook sheet `R_p1_mech`.

| quantity | value |
|---|---|
| hstar | 6.712 |
| q | 15.41 |
| pred | 0.8355, 0.7637, 0.6336, 0.3368, 0.1148, 0.2433, 0.08474, 0.02309 |
| loo | 0.8219, 0.7791, 0.63, 0.3715, 0.1108, 0.2334, 0.08459, 0.02309 |
| soils | S03, S04, S05, S06, S07, S08, S09, S10 |

## p1_modifiers.json

Source: `Manuscripts/results/p1_modifiers.json`; 20 values. Workbook sheet `R_p1_modifiers`.

| quantity | value |
|---|---|
| fP.mean | 0.4193 |
| fP.pos | 9 |
| fP.n | 9 |
| fP.p | 0.003906 |
| fT.mean | -0.1612 |
| fT.pos | 0 |
| fT.n | 10 |
| fT.p | 0.001953 |
| fh.mean | 0.1364 |
| fh.pos | 7 |
| fh.n | 10 |
| fh.p | 0.1602 |
| fD.mean | -0.04023 |
| fD.pos | 2 |
| fD.n | 10 |
| fD.p | 0.3223 |
| fN.mean | -0.008285 |
| fN.pos | 2 |
| fN.n | 10 |
| fN.p | 0.06445 |

## p1_rank.json

Source: `Manuscripts/results/p1_rank.json`; 20 values. Workbook sheet `R_p1_rank`.

| quantity | value |
|---|---|
| $n$ vs $/phi_D$.rho | 0.9515 |
| $n$ vs $/phi_D$.lo | 0.6883 |
| $n$ vs $/phi_D$.hi | 1 |
| $n$ vs $/phi_D$.p | 0.00015 |
| $/phi_D$ vs $/psi_D$(E16).rho | 0.8545 |
| $/phi_D$ vs $/psi_D$(E16).lo | 0.3703 |
| $/phi_D$ vs $/psi_D$(E16).hi | 1 |
| $/phi_D$ vs $/psi_D$(E16).p | 0.00195 |
| $/phi_D$ vs $/psi_D$(E7).rho | 0.8909 |
| $/phi_D$ vs $/psi_D$(E7).lo | 0.5472 |
| $/phi_D$ vs $/psi_D$(E7).hi | 1 |
| $/phi_D$ vs $/psi_D$(E7).p | 0.00135 |
| $n$ vs $/psi_D$(E16).rho | 0.697 |
| $n$ vs $/psi_D$(E16).lo | -0.0566 |
| $n$ vs $/psi_D$(E16).hi | 1 |
| $n$ vs $/psi_D$(E16).p | 0.03185 |
| $n$ vs $/psi_D$(E7).rho | 0.8909 |
| $n$ vs $/psi_D$(E7).lo | 0.4717 |
| $n$ vs $/psi_D$(E7).hi | 1 |
| $n$ vs $/psi_D$(E7).p | 0.001 |

## p1_rate_law.json

Source: `Manuscripts/results/p1_rate_law.json`; 17 values. Workbook sheet `R_p1_rate_law`.

| quantity | value |
|---|---|
| q_star_cm_d | 71.46 |
| R2 | 0.9715 |
| boot_ci95 | 58.38, 105.2 |
| n_boot | 4999 |
| jackknife_range | 65.83, 79.97 |
| loso_rmse_rate_law | 0.05811 |
| loso_8.mean_only | 0.3384 |
| loso_8.TAW | 0.08835 |
| loso_8.Se100 | 0.06407 |
| loso_8.n | 0.1004 |
| loso_8.log10Ks_linear | 0.06183 |
| loso_10.mean_only | 0.3178 |
| loso_10.TAW | 0.1825 |
| loso_10.Se100 | 0.1358 |
| loso_10.n | 0.09716 |

| quantity | value |
|---|---|
| loo_pred | 0.9116, 0.8418, 0.5702, 0.2502, 0.07804, 0.3136, 0.08007, 0.02297 |
| soils | S03, S04, S05, S06, S07, S08, S09, S10 |

## p1_table.csv

Source: `Manuscripts/results/p1_table.csv`; 10 rows. Workbook sheet `R_p1_table`.

| Unnamed: 0 | phiD_E9 | phiT_E9 | phiE_E9 | capFC | psiD_E16 | psiI_E16 | psiT_E16 | dE16 |
|---|---|---|---|---|---|---|---|---|
| S01 | 0.02912 | 0.75 | 0.1451 | 82.79 | 0.2946 | 0.6189 | 0.1113 | 218.5 |
| S02 | 0.265 | 0.4187 | 0.1927 | 104 | 0.193 | 0.6578 | 0.157 | 143.6 |
| S03 | 0.8455 | 0.07982 | 0.07403 | 4.172 | 1.21 | -0.3839 | 0.1785 | 99.64 |
| S04 | 0.7409 | 0.1535 | 0.09992 | 13.19 | 0.7993 | 0.2189 | -0.007351 | 115 |
| S05 | 0.643 | 0.2043 | 0.1277 | 41.33 | 0.5821 | 0.4146 | 0.01543 | 141.1 |
| S06 | 0.2892 | 0.5825 | 0.04019 | 131.8 | 0.469 | 0.4334 | 0.1068 | 191.5 |
| S07 | 0.1473 | 0.7155 | 0.04151 | 208.1 | 0.3625 | 0.5278 | 0.1168 | 217.6 |
| S08 | 0.2852 | 0.5547 | 0.07846 | 57.04 | 0.533 | 0.3857 | 0.09111 | 152.5 |
| S09 | 0.08702 | 0.7744 | 0.02717 | 167.5 | 0.3117 | 0.645 | 0.04777 | 217.4 |
| S10 | 0.02197 | 0.8416 | 0.02489 | 173 | 0.3103 | 0.838 | -0.01707 | 222.1 |

## p1_table2.csv

Source: `Manuscripts/results/p1_table2.csv`; 10 rows. Workbook sheet `R_p1_table2`.

| Unnamed: 0 | phiD_E9 | phiT_E9 | phiE_E9 | capFC | psiD_E16 | psiI_E16 | psiT_E16 | dE16 |
|---|---|---|---|---|---|---|---|---|
| S01 | 0.02912 | 0.75 | 0.1451 | 82.79 | 0.2946 | 0.6189 | 0.1113 | 218.5 |
| S02 | 0.265 | 0.4187 | 0.1927 | 104 | 0.193 | 0.6578 | 0.157 | 143.6 |
| S03 | 0.8455 | 0.07982 | 0.07403 | 4.172 | 1.21 | -0.3839 | 0.1785 | 99.64 |
| S04 | 0.7409 | 0.1535 | 0.09992 | 13.19 | 0.7993 | 0.2189 | -0.007351 | 115 |
| S05 | 0.643 | 0.2043 | 0.1277 | 41.33 | 0.5821 | 0.4146 | 0.01543 | 141.1 |
| S06 | 0.2892 | 0.5825 | 0.04019 | 131.8 | 0.469 | 0.4334 | 0.1068 | 191.5 |
| S07 | 0.1473 | 0.7155 | 0.04151 | 208.1 | 0.3625 | 0.5278 | 0.1168 | 217.6 |
| S08 | 0.2852 | 0.5547 | 0.07846 | 57.04 | 0.533 | 0.3857 | 0.09111 | 152.5 |
| S09 | 0.08702 | 0.7744 | 0.02717 | 167.5 | 0.3117 | 0.645 | 0.04777 | 217.4 |
| S10 | 0.02197 | 0.8416 | 0.02489 | 173 | 0.3103 | 0.838 | -0.01707 | 222.1 |

## p2_R.json

Source: `Manuscripts/results/p2_R.json`; 27 values. Workbook sheet `R_p2_R`.

| quantity | value |
|---|---|
| R_model.terms | lnI, n, cap, dT |

| quantity | value |
|---|---|
| R_model.coef.const | 3.355 |
| R_model.coef.lnI | 0.1944 |
| R_model.coef.n | 0.4917 |
| R_model.coef.cap | 0.1454 |
| R_model.coef.dT | 0.2383 |
| R_model.se.const | 0.02376 |
| R_model.se.lnI | 0.001882 |
| R_model.se.n | 0.01289 |
| R_model.se.cap | 0.004217 |
| R_model.se.dT | 0.00723 |
| R_model.r2 | 0.6111 |
| R_model.path[0] | lnI, 0.5509 |
| R_model.path[1] | n, 0.5356 |
| R_model.path[2] | cap, 0.5169 |
| R_model.path[3] | dT, 0.5137 |
| R_model.loeo_rmse_ln | 0.5137 |
| R_sparse.R2 | 0.3028 |
| R_sparse.rmse | 80.54 |
| R_sparse.R2ln | 0.4092 |
| R_GBM_all.R2 | 0.4956 |
| R_GBM_all.rmse | 68.51 |
| R_GBM_all.R2ln | 0.4268 |
| R_GBM_sel.R2 | 0.5095 |
| R_GBM_sel.rmse | 67.56 |
| R_GBM_sel.R2ln | 0.4665 |
| R_sparse_LOSO_R2ln | 0.4275 |

## p2_a.json

Source: `Manuscripts/results/p2_a.json`; 46 values. Workbook sheet `R_p2_a`.

| quantity | value |
|---|---|
| svd.sv | 240.3, 82.6, 57.95, 39.37, 21.84, 18.17, 5.461 |
| svd.null[0].k | 1 |
| svd.null[0].sv | 5.461 |
| svd.null[0].vec.rain_mm | 0.8956 |
| svd.null[0].vec.irrigation_mm | 0.8343 |
| svd.null[0].vec.ET_mm | -1 |
| svd.null[0].vec.transpiration_mm | 0.3675 |
| svd.null[0].vec.soil_evap_mm | 0.1207 |
| svd.null[0].vec.canopy_evap_mm | -0.1216 |
| svd.null[0].vec.drainage_1m_mm | -0.9325 |
| svd.null[1].k | 2 |
| svd.null[1].sv | 18.17 |
| svd.null[1].vec.rain_mm | -0.3585 |
| svd.null[1].vec.irrigation_mm | -0.2111 |

| quantity | value |
|---|---|
| svd.null[1].vec.ET_mm | -0.1925 |
| svd.null[1].vec.transpiration_mm | 0.7876 |
| svd.null[1].vec.soil_evap_mm | 0.4982 |
| svd.null[1].vec.canopy_evap_mm | 1 |
| svd.null[1].vec.drainage_1m_mm | 0.1799 |
| svd.null[2].k | 3 |
| svd.null[2].sv | 21.84 |
| svd.null[2].vec.rain_mm | -0.03927 |
| svd.null[2].vec.irrigation_mm | 0.03899 |
| svd.null[2].vec.ET_mm | -0.02946 |
| svd.null[2].vec.transpiration_mm | -0.3726 |
| svd.null[2].vec.soil_evap_mm | 0.2027 |
| svd.null[2].vec.canopy_evap_mm | 1 |
| svd.null[2].vec.drainage_1m_mm | -0.0192 |
| balance (W-ET-D) | 47.83, 5.435, 81.74 |
| ET partition (ET-T-Es-Ec) | 190, 129, 334.7 |
| clrpca | 0.8858, 0.07317, 0.03278, 0.008299, 0 |
| clr_load.transpiration_mm.0 | -0.218 |
| clr_load.transpiration_mm.1 | 0.035 |
| clr_load.transpiration_mm.2 | -0.22 |
| clr_load.soil_evap_mm.0 | -0.334 |
| clr_load.soil_evap_mm.1 | -0.557 |
| clr_load.soil_evap_mm.2 | -0.449 |
| clr_load.canopy_evap_mm.0 | -0.034 |
| clr_load.canopy_evap_mm.1 | 0.801 |
| clr_load.canopy_evap_mm.2 | -0.21 |
| clr_load.R.0 | -0.285 |
| clr_load.R.1 | -0.079 |
| clr_load.R.2 | 0.839 |
| clr_load.drainage_1m_mm.0 | 0.871 |
| clr_load.drainage_1m_mm.1 | -0.2 |
| clr_load.drainage_1m_mm.2 | 0.04 |

## p2_dpf.json

Source: `Manuscripts/results/p2_dpf.json`; 38 values. Workbook sheet `R_p2_dpf`.

| quantity | value |
|---|---|
| exogenous.terms | lnW/_TAW, lnalpha, Ifrac, dT |
| exogenous.hist[0] | lnW/_TAW, 0.1926 |
| exogenous.hist[1] | lnalpha, 0.2601 |
| exogenous.hist[2] | Ifrac, 0.422 |
| exogenous.hist[3] | dT, 0.4334 |
| exogenous.loeo | 0.4334 |
| exogenous.loso | 0.2171 |
| exogenous.r2 | 0.5752 |

| quantity | value |
|---|---|
| exogenous.coef.const | -12.1 |
| exogenous.coef.lnW_TAW | 4.411 |
| exogenous.coef.lnalpha | -1.214 |
| exogenous.coef.Ifrac | -5.565 |
| exogenous.coef.dT | -0.9978 |
| exogenous.se.const | 0.1018 |
| exogenous.se.lnW_TAW | 0.03957 |
| exogenous.se.lnalpha | 0.0273 |
| exogenous.se.Ifrac | 0.1049 |
| exogenous.se.dT | 0.03043 |
| exogenous.gbm_loeo | 0.4975 |
| with_partition.terms | lnW/_TAW, lnalpha, Ifrac, dT |
| with_partition.hist[0] | lnW/_TAW, 0.1926 |
| with_partition.hist[1] | lnalpha, 0.2601 |
| with_partition.hist[2] | Ifrac, 0.422 |
| with_partition.hist[3] | dT, 0.4334 |
| with_partition.loeo | 0.4334 |
| with_partition.loso | 0.2171 |
| with_partition.r2 | 0.5752 |
| with_partition.coef.const | -12.1 |
| with_partition.coef.lnW_TAW | 4.411 |
| with_partition.coef.lnalpha | -1.214 |
| with_partition.coef.Ifrac | -5.565 |
| with_partition.coef.dT | -0.9978 |
| with_partition.se.const | 0.1018 |
| with_partition.se.lnW_TAW | 0.03957 |
| with_partition.se.lnalpha | 0.0273 |
| with_partition.se.Ifrac | 0.1049 |
| with_partition.se.dT | 0.03043 |
| with_partition.gbm_loeo | 0.5906 |

## p2_et.json

Source: `Manuscripts/results/p2_et.json`; 90 values. Workbook sheet `R_p2_et`.

| quantity | value |
|---|---|
| lnET.terms | lnW, n, lnI1 |
| lnET.hist[0] | lnW, 0.8008 |
| lnET.hist[1] | n, 0.8947 |
| lnET.hist[2] | lnI1, 0.8982 |
| lnET.loeo | 0.8982 |
| lnET.loso | 0.9037 |
| lnET.r2 | 0.9082 |
| lnET.coef.const | 2.503 |
| lnET.coef.lnW | 0.6458 |
| lnET.coef.n | -0.2992 |

| quantity | value |
|---|---|
| lnET.coef.lnI1 | 0.02932 |
| lnET.se.const | 0.02691 |
| lnET.se.lnW | 0.005084 |
| lnET.se.n | 0.004524 |
| lnET.se.lnI1 | 0.001125 |
| lnET.gbm_loeo | 0.8789 |
| lnET.gbm_loso | 0.8936 |
| lnET.rmse_loeo | 0.143 |
| lnT.terms | lnW, n |
| lnT.hist[0] | lnW, 0.5805 |
| lnT.hist[1] | n, 0.6409 |
| lnT.loeo | 0.6409 |
| lnT.loso | 0.6524 |
| lnT.r2 | 0.6712 |
| lnT.coef.const | -1.329 |
| lnT.coef.lnW | 1.127 |
| lnT.coef.n | -0.5007 |
| lnT.se.const | 0.06105 |
| lnT.se.lnW | 0.00874 |
| lnT.se.n | 0.01243 |
| lnT.gbm_loeo | 0.6415 |
| lnT.gbm_loso | 0.6592 |
| lnT.rmse_loeo | 0.4588 |
| lnEs.terms | n, mulch, lnrain |
| lnEs.hist[0] | n, 0.07031 |
| lnEs.hist[1] | mulch, 0.2369 |
| lnEs.hist[2] | lnrain, 0.3084 |
| lnEs.loeo | 0.3084 |
| lnEs.loso | 0.3484 |
| lnEs.r2 | 0.4363 |
| lnEs.coef.const | 3.271 |
| lnEs.coef.n | -0.7698 |
| lnEs.coef.mulch | -0.3804 |
| lnEs.coef.lnrain | 0.4616 |
| lnEs.se.const | 0.06683 |
| lnEs.se.n | 0.01217 |
| lnEs.se.mulch | 0.006953 |
| lnEs.se.lnrain | 0.01089 |
| lnEs.gbm_loeo | 0.5468 |
| lnEs.gbm_loso | 0.4758 |
| lnEs.rmse_loeo | 0.4803 |
| lnEc.terms | lnW, lnrain, Ifrac |
| lnEc.hist[0] | lnW, 0.6408 |
| lnEc.hist[1] | lnrain, 0.6513 |
| lnEc.hist[2] | Ifrac, 0.6677 |
| lnEc.loeo | 0.6677 |
| lnEc.loso | 0.6695 |
| lnEc.r2 | 0.7035 |

| quantity | value |
|---|---|
| lnEc.coef.const | -13.32 |
| lnEc.coef.lnW | -0.4701 |
| lnEc.coef.lnrain | 3.157 |
| lnEc.coef.Ifrac | 4.24 |
| lnEc.se.const | 0.1448 |
| lnEc.se.lnW | 0.1351 |
| lnEc.se.lnrain | 0.1263 |
| lnEc.se.Ifrac | 0.2092 |
| lnEc.gbm_loeo | 0.6914 |
| lnEc.gbm_loso | 0.8059 |
| lnEc.rmse_loeo | 0.9389 |
| lnY.terms | lnW, lnI1, lnrain, mulch |
| lnY.hist[0] | lnW, 0.5314 |
| lnY.hist[1] | lnI1, 0.5492 |
| lnY.hist[2] | lnrain, 0.5992 |
| lnY.hist[3] | mulch, 0.604 |
| lnY.loeo | 0.604 |
| lnY.loso | 0.6169 |
| lnY.r2 | 0.6354 |
| lnY.coef.const | -9.681 |
| lnY.coef.lnW | -0.162 |
| lnY.coef.lnI1 | 0.3226 |
| lnY.coef.lnrain | 1.716 |
| lnY.coef.mulch | 0.1489 |
| lnY.se.const | 0.1806 |
| lnY.se.lnW | 0.05911 |
| lnY.se.lnI1 | 0.008161 |
| lnY.se.lnrain | 0.04756 |
| lnY.se.mulch | 0.01476 |
| lnY.gbm_loeo | 0.6501 |
| lnY.gbm_loso | 0.7425 |
| lnY.rmse_loeo | 0.9531 |

## p2_etlaw.json

Source: `Manuscripts/results/p2_etlaw.json`; 31 values. Workbook sheet `R_p2_etlaw`.

| quantity | value |
|---|---|
| sparse_loeo | 0.8982 |
| sparse_loso | 0.9037 |
| gbm_loeo | 0.876 |
| gbm_loso | 0.8812 |
| coef_mean | 2.508, 0.6448, -0.2984, 0.02945 |
| coef_sd | 0.0771, 0.01676, 0.01733, 0.002268 |
| coefS_mean | 2.498, 0.6465, -0.2993, 0.02912 |
| coefS_sd | 0.07797, 0.0124, 0.007435, 0.002257 |

| quantity | value |
|---|---|
| rmse_mm_loeo | 68.82 |
| mape_loeo | 9.179 |
| by_exp[0] | E0/_Baselines, 10, 5.834, -25.17 |
| by_exp[1] | E10/_Amount/_Timing/_Soil, 954, 3.256, 1.285 |
| by_exp[2] | E11/_Temp/_Evap/_Rain, 1000, 8.999, -62.58 |
| by_exp[3] | E12/_Sensor/_Trigger/_Rain, 1000, 3.268, -15.24 |
| by_exp[4] | E13/_N/_Water/_Soil, 975, 3.532, -3.65 |
| by_exp[5] | E14/_Mulch/_Rain, 1000, 4.024, 7.484 |
| by_exp[6] | E15/_Method/_Depth/_Amount, 1000, 4.978, 28.96 |
| by_exp[7] | E16/_Edge/_Cases, 740, 16.6, 34.49 |
| by_exp[8] | E1/_Doses, 189, 3.76, 9.833 |
| by_exp[9] | E2/_TriggerPressure, 197, 3.002, -6.6 |
| by_exp[10] | E3/_SensorDepth, 200, 2.648, -8.047 |
| by_exp[11] | E4/_Methods, 96, 3.557, 9.425 |
| by_exp[12] | E5/_SHP/_Dynamics, 300, 11.7, -63.8 |
| by_exp[13] | E6/_Mediterranean/_Pollution, 193, 8.457, 41.52 |
| by_exp[14] | E7/_Mulching, 200, 4.66, 10.62 |
| by_exp[15] | E8/_Rain/_Intensity/_Moisture, 1000, 16.72, -11.34 |
| by_exp[16] | E9/_Soil/_Moisture/_Rain, 987, 27.4, 12.89 |
| closureD.r2 | 0.6877 |
| closureD.rmse | 63.71 |
| closureD.n | 8054 |
| closureD.negshare | 0.04817 |

## p2_laws.json

Source: `Manuscripts/results/p2_laws.json`; 12 values. Workbook sheet `R_p2_laws`.

| quantity | value |
|---|---|
| svd4.vec | 1.037, 0.9575, -1, -1.036 |
| svd4.lo | 1.031, 0.9396, -1, -1.052 |
| svd4.hi | 1.045, 0.9738, -1, -1.023 |
| svd4.sv | 176.6, 75.65, 56.89, 5.92 |
| svd4_perm_minSV | 33.33 |
| svd4_noE89.vec | 1.021, 0.9688, -1, -1.019 |
| svd4_noE89.sv | 158.9, 67.04, 49.48, 3.007 |
| closure.all_median_abs | 0.009134 |
| closure.p95 | 0.3515 |
| closure.noE89_median | 0.006472 |
| closure.noE89_p95 | 0.02268 |
| closure.E89_median | 0.1643 |

## p2_laws2.json

Source: `Manuscripts/results/p2_laws2.json`; 81 values. Workbook sheet `R_p2_laws2`.

| quantity | value |
|---|---|
| comp_lattice[0].exp | E11_Temp_Evap_Rain |
| comp_lattice[0].n | 1000 |
| comp_lattice[0].irrigation_mm_slope | -0.4193 |
| comp_lattice[0].irrigation_mm_se | 0.01881 |
| comp_lattice[0].ET_mm_slope | 0.4302 |
| comp_lattice[0].ET_mm_se | 0.02525 |
| comp_lattice[0].drainage_1m_mm_slope | 0.1333 |
| comp_lattice[0].drainage_1m_mm_se | 0.006955 |
| comp_lattice[0].W_slope | 0.5807 |
| comp_lattice[0].W_se | 0.01881 |
| comp_lattice[1].exp | E12_Sensor_Trigger_Rain |
| comp_lattice[1].n | 1000 |
| comp_lattice[1].irrigation_mm_slope | -0.673 |
| comp_lattice[1].irrigation_mm_se | 0.01357 |
| comp_lattice[1].ET_mm_slope | 0.07769 |
| comp_lattice[1].ET_mm_se | 0.009059 |
| comp_lattice[1].drainage_1m_mm_slope | 0.2316 |
| comp_lattice[1].drainage_1m_mm_se | 0.006128 |
| comp_lattice[1].W_slope | 0.327 |
| comp_lattice[1].W_se | 0.01357 |
| comp_lattice[2].exp | E14_Mulch_Rain |
| comp_lattice[2].n | 1000 |
| comp_lattice[2].irrigation_mm_slope | -0.6656 |
| comp_lattice[2].irrigation_mm_se | 0.006611 |
| comp_lattice[2].ET_mm_slope | -0.01901 |
| comp_lattice[2].ET_mm_se | 0.009262 |
| comp_lattice[2].drainage_1m_mm_slope | 0.3204 |
| comp_lattice[2].drainage_1m_mm_se | 0.005513 |
| comp_lattice[2].W_slope | 0.3344 |
| comp_lattice[2].W_se | 0.006611 |
| comp16.S01.kappa | 0.3033 |
| comp16.S01.dET | 0.5118 |
| comp16.S01.dD | 0.186 |
| comp16.S01.n | 32 |
| comp16.S02.kappa | 0.1641 |
| comp16.S02.dET | 0.346 |
| comp16.S02.dD | 0.4954 |
| comp16.S02.n | 8 |
| comp16.S03.kappa | -0.0141 |
| comp16.S03.dET | 0.4732 |
| comp16.S03.dD | 0.5502 |
| comp16.S03.n | 32 |
| comp16.S04.kappa | 0.08366 |

| quantity | value |
|---|---|
| comp16.S04.dET | 0.4733 |
| comp16.S04.dD | 0.4532 |
| comp16.S04.n | 32 |
| comp16.S05.kappa | 0.1021 |
| comp16.S05.dET | 0.5515 |
| comp16.S05.dD | 0.3576 |
| comp16.S05.n | 32 |
| comp16.S06.kappa | 0.1082 |
| comp16.S06.dET | 0.6193 |
| comp16.S06.dD | 0.2813 |
| comp16.S06.n | 32 |
| comp16.S07.kappa | 0.2158 |
| comp16.S07.dET | 0.5423 |
| comp16.S07.dD | 0.2464 |
| comp16.S07.n | 32 |
| comp16.S08.kappa | 0.08991 |
| comp16.S08.dET | 0.6131 |
| comp16.S08.dD | 0.3076 |
| comp16.S08.n | 32 |
| comp16.S09.kappa | 0.2382 |
| comp16.S09.dET | 0.5154 |
| comp16.S09.dD | 0.2529 |
| comp16.S09.n | 32 |
| comp16.S10.kappa | 0.3016 |
| comp16.S10.dET | 0.4768 |
| comp16.S10.dD | 0.2369 |
| comp16.S10.n | 32 |
| kappa16_pool.est | 0.1589 |
| kappa16_pool.lo | 0.1336 |
| kappa16_pool.hi | 0.1813 |
| kap_by.fEp.0.3 | 0.1927 |
| kap_by.fEp.1.2 | 0.1232 |
| kap_by.fT.2.0 | 0.188 |
| kap_by.fT.6.0 | 0.1289 |
| kap_by.fh.50.0 | 0.211 |
| kap_by.fh.2000.0 | 0.1067 |
| kap_by.fD.1.2 | 0.0791 |
| kap_by.fD.30.0 | 0.2408 |

## p2_nested.json

Source: `Manuscripts/results/p2_nested.json`; 18 values. Workbook sheet `R_p2_nested`.

| quantity | value |
|---|---|
| r2 | 0.8946 |
| sets[0] | lnW, n, lnI1 |

| quantity | value |
|---|---|
| sets[1] | lnW, n, lnI1 |
| sets[2] | lnW, n, lnI1 |
| sets[3] | lnW, n, lnI1 |
| sets[4] | lnW, n, lnI1 |
| sets[5] | lnW, n, lnI1 |
| sets[6] | lnW, n, lnI1 |
| sets[7] | lnW, n |
| sets[8] | lnW, n, lnI1 |
| sets[9] | lnW, n, lnI1 |
| sets[10] | lnW, n, lnI1 |
| sets[11] | lnW, n, lnI1 |
| sets[12] | lnW, n, lnI1 |
| sets[13] | lnW, n |
| sets[14] | lnW, n, lnI1 |
| sets[15] | lnW, n, lnI1 |
| sets[16] | lnW, n, lnI1 |

## p2_pca.json

Source: `Manuscripts/results/p2_pca.json`; 4 values. Workbook sheet `R_p2_pca`.

| quantity | value |
|---|---|
| evr | 0.8858, 0.07317, 0.03278 |
| load[0] | -0.2184, -0.3338, -0.03369, -0.2851, 0.8709 |
| load[1] | 0.03458, -0.5574, 0.8013, -0.07872, -0.1997 |
| load[2] | -0.2197, -0.4488, -0.2104, 0.8393, 0.03956 |

## p2_pca_sens.json

Source: `Manuscripts/results/p2_pca_sens.json`; 10 values. Workbook sheet `R_p2_pca_sens`.

| quantity | value |
|---|---|
| sens.0.001 | 0.8858, 0.07317, 0.03278, 0.008299 |
| sens.0.1 | 0.7435, 0.1732, 0.06629, 0.01701 |
| sens.1 | 0.6274, 0.2481, 0.09837, 0.02614 |
| sens.5 | 0.565, 0.2511, 0.1481, 0.03581 |
| sens.10 | 0.5233, 0.271, 0.168, 0.03776 |
| clip1_evr | 0.6274, 0.2481, 0.09837 |
| clip1_load[0] | -0.1982, -0.3672, 0.06212, -0.338, 0.8413 |
| clip1_load[1] | 0.08674, -0.5565, 0.7747, -0.01804, -0.2869 |
| clip1_load[2] | -0.2647, -0.4327, -0.2259, 0.826, 0.09732 |
| cor_PC1 | 0.4828, 0.454, 0.9791 |

## p2_robust.json

Source: `Manuscripts/results/p2_robust.json`; 121 values. Workbook sheet `R_p2_robust`.

| quantity | value |
|---|---|
| terms | lnW, n, lnI1 |
| bootstrap.n_resamples | 2000 |
| bootstrap.law_loeo | 0.8982 |
| bootstrap.gbm3_loeo | 0.8662 |
| bootstrap.diff | 0.03198 |
| bootstrap.diff_mean | 0.03612 |
| bootstrap.ci95 | 0.004383, 0.0867 |
| without_E8_E9.n_runs | 8054 |
| without_E8_E9.n_experiments | 15 |
| without_E8_E9.loeo | 0.8818 |
| without_E8_E9.loso | 0.8811 |
| without_E8_E9.rmse_mm | 65.88 |
| without_E8_E9.mape_pct | 5.787 |
| without_E8_E9.steps[0].terms | lnW |
| without_E8_E9.steps[0].loeo | 0.5998 |
| without_E8_E9.steps[0].rmse_mm | 137.9 |
| without_E8_E9.steps[1].terms | lnW, n |
| without_E8_E9.steps[1].loeo | 0.8567 |
| without_E8_E9.steps[1].rmse_mm | 73.84 |
| without_E8_E9.steps[2].terms | lnW, n, lnI1 |
| without_E8_E9.steps[2].loeo | 0.8818 |
| without_E8_E9.steps[2].rmse_mm | 65.88 |
| without_E8_E9.gbm3_loeo | 0.7781 |
| table1.rain_mm.mean | 417 |
| table1.rain_mm.sd | 143.3 |
| table1.rain_mm.q25 | 297.4 |
| table1.rain_mm.median | 464.4 |
| table1.rain_mm.q75 | 464.4 |
| table1.irrigation_mm.mean | 179.7 |
| table1.irrigation_mm.sd | 191.4 |
| table1.irrigation_mm.q25 | 0 |
| table1.irrigation_mm.median | 178.2 |
| table1.irrigation_mm.q75 | 268.4 |
| table1.W.mean | 596.7 |
| table1.W.sd | 251 |
| table1.W.q25 | 464.4 |
| table1.W.median | 654.5 |
| table1.W.q75 | 722.1 |
| table1.ET_mm.mean | 546.1 |
| table1.ET_mm.sd | 195.7 |
| table1.ET_mm.q25 | 385.9 |
| table1.ET_mm.median | 611.5 |
| table1.ET_mm.q75 | 670.7 |

| quantity | value |
|---|---|
| table1.drainage_1m_mm.mean | 60.09 |
| table1.drainage_1m_mm.sd | 105.8 |
| table1.drainage_1m_mm.q25 | 2.803 |
| table1.drainage_1m_mm.median | 33.51 |
| table1.drainage_1m_mm.q75 | 77.62 |
| table1.R.mean | 163.7 |
| table1.R.sd | 96.47 |
| table1.R.q25 | 102.8 |
| table1.R.median | 129 |
| table1.R.q75 | 222.5 |
| R_by_experiment.E0_Baselines.runs | 10 |
| R_by_experiment.E0_Baselines.R_median | 106.9 |
| R_by_experiment.E0_Baselines.R_ET_median | 0.2398 |
| R_by_experiment.E0_Baselines.dS_W_median | -0.02058 |
| R_by_experiment.E10_Amount_Timing_Soil.runs | 954 |
| R_by_experiment.E10_Amount_Timing_Soil.R_median | 118.3 |
| R_by_experiment.E10_Amount_Timing_Soil.R_ET_median | 0.2295 |
| R_by_experiment.E10_Amount_Timing_Soil.dS_W_median | -0.01078 |
| R_by_experiment.E11_Temp_Evap_Rain.runs | 1000 |
| R_by_experiment.E11_Temp_Evap_Rain.R_median | 296.3 |
| R_by_experiment.E11_Temp_Evap_Rain.R_ET_median | 0.3751 |
| R_by_experiment.E11_Temp_Evap_Rain.dS_W_median | -0.009339 |
| R_by_experiment.E12_Sensor_Trigger_Rain.runs | 1000 |
| R_by_experiment.E12_Sensor_Trigger_Rain.R_median | 217.6 |
| R_by_experiment.E12_Sensor_Trigger_Rain.R_ET_median | 0.3186 |
| R_by_experiment.E12_Sensor_Trigger_Rain.dS_W_median | -0.008244 |
| R_by_experiment.E13_N_Water_Soil.runs | 975 |
| R_by_experiment.E13_N_Water_Soil.R_median | 124.5 |
| R_by_experiment.E13_N_Water_Soil.R_ET_median | 0.284 |
| R_by_experiment.E13_N_Water_Soil.dS_W_median | -0.006363 |
| R_by_experiment.E14_Mulch_Rain.runs | 1000 |
| R_by_experiment.E14_Mulch_Rain.R_median | 262.1 |
| R_by_experiment.E14_Mulch_Rain.R_ET_median | 0.409 |
| R_by_experiment.E14_Mulch_Rain.dS_W_median | 0.0008958 |
| R_by_experiment.E15_Method_Depth_Amount.runs | 1000 |
| R_by_experiment.E15_Method_Depth_Amount.R_median | 103.2 |
| R_by_experiment.E15_Method_Depth_Amount.R_ET_median | 0.1689 |
| R_by_experiment.E15_Method_Depth_Amount.dS_W_median | 0.003355 |
| R_by_experiment.E16_Edge_Cases.runs | 740 |
| R_by_experiment.E16_Edge_Cases.R_median | 157 |
| R_by_experiment.E16_Edge_Cases.R_ET_median | 0.2964 |
| R_by_experiment.E16_Edge_Cases.dS_W_median | -0.01008 |
| R_by_experiment.E1_Doses.runs | 189 |
| R_by_experiment.E1_Doses.R_median | 117.6 |
| R_by_experiment.E1_Doses.R_ET_median | 0.2123 |
| R_by_experiment.E1_Doses.dS_W_median | -0.001241 |
| R_by_experiment.E2_TriggerPressure.runs | 197 |
| R_by_experiment.E2_TriggerPressure.R_median | 167.5 |

| quantity | value |
|---|---|
| R_by_experiment.E2_TriggerPressure.R_ET_median | 0.2979 |
| R_by_experiment.E2_TriggerPressure.dS_W_median | -0.008657 |
| R_by_experiment.E3_SensorDepth.runs | 200 |
| R_by_experiment.E3_SensorDepth.R_median | 201.8 |
| R_by_experiment.E3_SensorDepth.R_ET_median | 0.3213 |
| R_by_experiment.E3_SensorDepth.dS_W_median | -0.008657 |
| R_by_experiment.E4_Methods.runs | 96 |
| R_by_experiment.E4_Methods.R_median | 103.4 |
| R_by_experiment.E4_Methods.R_ET_median | 0.1863 |
| R_by_experiment.E4_Methods.dS_W_median | -0.003341 |
| R_by_experiment.E5_SHP_Dynamics.runs | 300 |
| R_by_experiment.E5_SHP_Dynamics.R_median | 161.5 |
| R_by_experiment.E5_SHP_Dynamics.R_ET_median | 0.2882 |
| R_by_experiment.E5_SHP_Dynamics.dS_W_median | -0.007413 |
| R_by_experiment.E6_Mediterranean_Pollution.runs | 193 |
| R_by_experiment.E6_Mediterranean_Pollution.R_median | 70.43 |
| R_by_experiment.E6_Mediterranean_Pollution.R_ET_median | 0.1435 |
| R_by_experiment.E6_Mediterranean_Pollution.dS_W_median | -0.01191 |
| R_by_experiment.E7_Mulching.runs | 200 |
| R_by_experiment.E7_Mulching.R_median | 226.3 |
| R_by_experiment.E7_Mulching.R_ET_median | 0.4091 |
| R_by_experiment.E7_Mulching.dS_W_median | -0.005667 |
| R_by_experiment.E8_Rain_Intensity_Moisture.runs | 1000 |
| R_by_experiment.E8_Rain_Intensity_Moisture.R_median | 39.29 |
| R_by_experiment.E8_Rain_Intensity_Moisture.R_ET_median | 0.1438 |
| R_by_experiment.E8_Rain_Intensity_Moisture.dS_W_median | -0.1335 |
| R_by_experiment.E9_Soil_Moisture_Rain.runs | 987 |
| R_by_experiment.E9_Soil_Moisture_Rain.R_median | 81.23 |
| R_by_experiment.E9_Soil_Moisture_Rain.R_ET_median | 0.3736 |
| R_by_experiment.E9_Soil_Moisture_Rain.dS_W_median | -0.01399 |

## p2_s02_excl.json

Source: `Manuscripts/results/p2_s02_excl.json`; 27 values. Workbook sheet `R_p2_s02_excl`.

| quantity | value |
|---|---|
| terms | lnW, n, lnI1 |
| full.loeo | 0.8982 |
| full.loso | 0.9037 |
| full.coef.const | 2.503 |
| full.coef.lnW | 0.6458 |
| full.coef.n | -0.2992 |
| full.coef.lnI1 | 0.02932 |
| full.R_mean | 163.7 |
| full.R_share_of_ET | 0.2998 |
| without_S02.n_runs | 9651 |

| quantity | value |
|---|---|
| without_S02.loeo | 0.8972 |
| without_S02.loso | 0.9037 |
| without_S02.coef.const | 2.461 |
| without_S02.coef.lnW | 0.6519 |
| without_S02.coef.n | -0.2972 |
| without_S02.coef.lnI1 | 0.02947 |
| without_S02.R_mean | 163.1 |
| without_S02.R_share_of_ET | 0.2988 |
| without_S02_S03.n_runs | 9145 |
| without_S02_S03.loeo | 0.8931 |
| without_S02_S03.loso | 0.8976 |
| without_S02_S03.coef.const | 2.548 |
| without_S02_S03.coef.lnW | 0.6419 |
| without_S02_S03.coef.n | -0.3168 |
| without_S02_S03.coef.lnI1 | 0.03039 |
| without_S02_S03.R_mean | 165.4 |
| without_S02_S03.R_share_of_ET | 0.2958 |

# S7. Where every table and figure comes from

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 1 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Figure 1 | Figure drawn by the script from the input data | Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p2_figs.py; Manuscripts/reproducibility/analysis/p2_fig1_new.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig01_pipeline_A4.pdf |
| Table 2 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Table 3 | Table typed from the result file (27 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 2 | Figure drawn by the script from the input data | Manuscripts/reproducibility/make_a4.py; Manuscripts/reproducibility/analysis/p1_figs_d.py; Manuscripts/reproducibility/analysis/p1_fig1_new.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig01_framework_A4.pdf |
| Figure 3 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_a.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig02_hydraulics.pdf |
| Table 4 | Table written by the script (tex/data_table.tex) | Manuscripts/reproducibility/analysis/data_overview.py | Data/Comprehensive_Summary/00_summary_all_simulations |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Figure 4 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/data_overview.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/data/fig_data_overview.pdf |
| Table 5 | Table typed in the manuscript from the result files of the script | p2_robust.py; p2_robust.js | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 6 | Table typed from the result file (11 of its numbers found there) | Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json |
| Figure 5 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figfail.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/figS_failed.pdf |
| Table 7 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Figure 6 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_a.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig03_memory_grid.pdf |
| Figure 7 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig04_fates.pdf |
| Table 8 | Table typed from the result file (55 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p1_core.py; p1_core.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 8 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig05_predictors.pdf |
| Figure 9 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_b.py; p1_rate_law.py; p1_rate_law.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig06_mechanism.pdf |
| Table 9 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p1_rate_law.py; p1_rate_law.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 10 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig08_replication.pdf |
| Figure 11 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_d.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig10_robustness.pdf |
| Table 10 | Table typed from the result file (14 of its numbers found there) | p1_stats.py; p1_figs_d.py; p1_rank.js | Data/daisy_ensemble.csv.gz (10 190 runs); Results/07_Derived_Data_Tables/pareto_runs.csv |
| Figure 12 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig07_modifiers.pdf |
| Table 11 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/analysis/p1_figs_c.py; p1_modifiers.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_modifiers.json |
| Figure 13 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_e.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig11_e11_surface.pdf |
| Table 12 | Table typed from the result file (7 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Figure 14 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs.py; p2_a.py; p2_laws.py; p2_a.js; p2_laws.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig02_invariants.pdf |
| Table 13 | Table typed from the result file (34 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Figure 15 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs.py; p2_R.py; p2_R.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig03_hidden_term.pdf |
| Table 14 | Table typed from the result file (22 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p1_core.py; p2_robust.py; p2_robust.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_core.json |
| Figure 16 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig4.py; Manuscripts/reproducibility/analysis/p2_figs.py; p2_a.py; p2_pca.js; p2_pca_sens.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig04_manifold.pdf |
| Table 15 | Table typed in the manuscript from the result files of the script | p2_a.py; p2_figs.py; p2_fig4.py; p2_pca.js; p2_pca_sens.js | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Figure 17 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig7.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig07_fates_by_soil.pdf |
| Figure 18 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_figs2.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig05_ET_law.pdf |
| Table 16 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 17 | Table typed from the result file (18 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json |
| Table 18 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 19 | Table typed from the result file (12 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Table 20 | Table typed from the result file (3 of its numbers found there) | Manuscripts/reproducibility/analysis/p4_reach.py; Manuscripts/reproducibility/analysis/-make_supplement_data.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p4_tableA1_reach.csv |
| Table 21 | Table typed from the result file (23 of its numbers found there) | Manuscripts/reproducibility/p3_js/run_drip.js; Manuscripts/reproducibility/analysis/p3_lit_figs.py; Manuscripts/reproducibility/analysis/make_supplement_data.py; p2_et.py; p2_dpf.py; p2_figs2.py; p2_nested.py; p2_et.js; p2_dpf.js; p2_etlaw.js; p2_nested.js; p2_s02_excl.py; p2_s02_excl.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/drip_out.json |
| Table 22 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Figure 19 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p2_fig6.py; p2_laws2.py; p2_laws2.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p2/fig06_compensation.pdf |
| Table 23 | Table typed from the result file (19 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/analysis/p2_fig6.py; Manuscripts/reproducibility/analysis/p2_laws2.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p2_laws2.json |
| Table 24 | Table typed from the result file (24 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/make_e12.py; Manuscripts/reproducibility/audit_numbers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12.json |
| Figure 20 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_fig14_ensemble_structure.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig14_ensemble_structure.pdf |
| Figure 21 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_fig13_time_management.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig13_time_management.pdf |
| Figure 22 | Figure drawn by the script from the input data | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/analysis/p1_fig12_drivers.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig12_drivers.pdf |
| Table 25 | Descriptive table (definitions, design, parameters or tests); no computation | - | |
| Table 26 | Table typed from the result file (8 of its numbers found there) | Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p3_figs.py; Manuscripts/reproducibility/p3_js/e1_heldout.js | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/e12_pred.json |
| Figure 23 | Figure drawn by the script from the input data | Manuscripts/reproducibility/fig_vg_check.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig15_vg_check.pdf |
| Figure 24 | Screen capture of SWAFL Desktop 1.1.0 (not regenerated) | - | Manuscripts/figures/shots_jpg/real_evap_pore.jpg; Manuscripts/figures/shots_jpg/real_evap_flux_cells.jpg; Manuscripts/figures/shots_jpg/real_tillage_micro.jpg |
| Figure 25 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p1_figs_c.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/figures/p1/fig09_recoverable.pdf |
| Table 27 | Table typed in the manuscript from the result files of the script | (see the reproducibility appendix of the paper) | Data/daisy_ensemble.csv.gz (10 190 runs) |
| Table 28 | Table typed from the result file (34 of its numbers found there) | Manuscripts/reproducibility/run_all.sh; Manuscripts/reproducibility/audit_numbers.py; Manuscripts/reproducibility/analysis/p1_core.py | Data/daisy_ensemble.csv.gz (10 190 runs); Manuscripts/results/p1_core.json |

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